Separation of Church and State Curricula? Public Standards, Private Values, and Textbook Content

Anjali Adukia and Emileigh Harrison

Working Paper, 2026

Prepared for ISEG Summer School 2026, by Benjamin W. Arold

Motivation & Research Question

- Curricula are a critical site of cultural transmission: they convey knowledge, instill values, and shape children's worldviews - with causal effects on later beliefs, religiosity, and careers.
- U.S. school-choice programs increasingly direct tax dollars to religious private schools and homeschooling, which face minimal curricular oversight - yet we know little about WHAT they teach.
- Question: How do the people, topics, and values in textbooks differ across public (Texas, California) vs. religious private/home-school settings, and over time?
  - Two puzzles: Why are 'red' Texas and 'blue' California textbooks so SIMILAR, while religious books DIVERGE - despite the polarization narrative?
  - New in this version: a market/regulatory framework that explains this pattern, tested via cross-state spillovers from standards revisions.

Data: A Novel Panel of 261 Textbooks (1980-2022)

- Three collections: Texas public, California public, and Religious (conservative Christian publishers Abeka, BJU Press, and ACE - widely used in private and home schools).
- Subjects: Reading, Science, Social Studies; grades 3 and 5 (primary-school-age children).
- Scale: 261 digitized textbooks, ~88,000 pages, ~16 million words, ~60,000 detected faces.
- Longitudinal, 1980-2022: enables comparisons across settings AND over four decades.
- Texas and California are the two largest K-12 markets and the conservative/liberal poles of public education; religious publishers serve over a third of taxpayer-funded non-Catholic Christian schools.

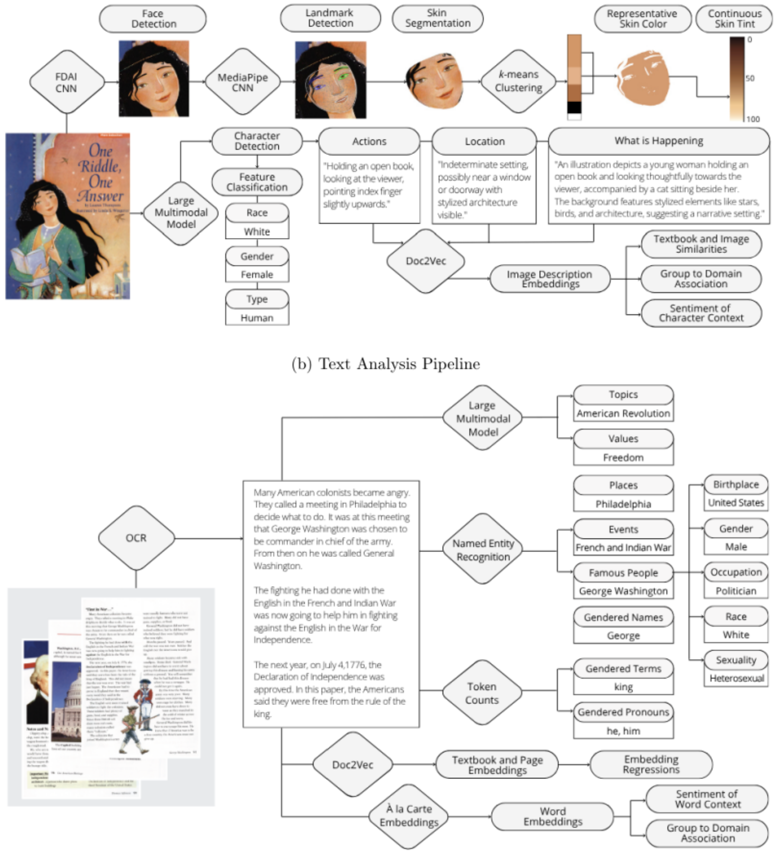
Computational Methods: What, Who, and How

- Each scanned page is treated as BOTH an image and a source of text, analyzed with AI + computational social-science tools at scale.
- WHAT is communicated: topics and values on each page (LLM + keyword validation).
- WHO is depicted: representation by race, gender, skin color, sexuality, birthplace, occupation (computer vision + NER).
- HOW people are portrayed: the roles, settings, actions, and sentiment surrounding each group (embeddings + VAD).
- Lets one study measure both PRESENCE and PORTRAYAL across settings and decades - replicating prior qualitative work while uncovering new patterns.

Method Pipeline: Turning Images and Text into Data

Figure 1: Data Extraction Pipelines: Image and Text Analysis

(a) Image Analysis Pipeline



Note: Image pipeline (top): a Large Multimodal Model (Gemini 2.5 Pro) detects characters, actions, setting, race/gender/type; a CNN face detector (FDAI) + MediaPipe landmarks feed skin segmentation and k-means clustering to a continuous skin tint (0=darkest, 100=lightest); Doc2Vec embeds image descriptions. Text pipeline (bottom): OCR feeds a GPT-4 LLM (topics/values), Named Entity Recognition (places, events, famous people), token counts of gendered terms, and page/word (a-la-carte) embeddings. (Figure 1)

Computational Methods in Depth

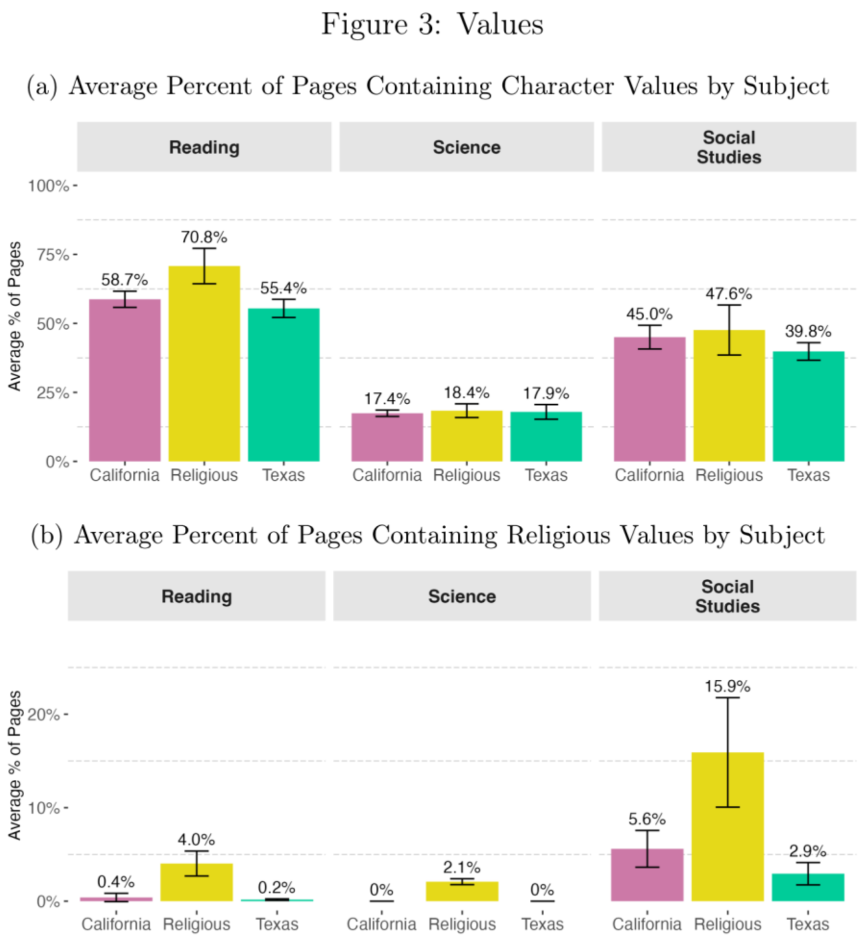
- Images (LMM): Gemini 2.5 Pro detects characters (incl. faces not visible & non-human), their race, gender, actions and setting; outperforms the CNN classifier used for robustness.
- Skin color (computer vision): CNN-detected faces are skin-segmented, k-means clustered, and scored on the L* axis of CIE L*a*b* space (0=darkest to 100=lightest).
- LLM (GPT-4): free-form annotation of topics and values per page (hypothesis-generating, not a fixed list); validated vs. human labels and keyword counts.
- NER + Pantheon 2.0: identify places, events, and famous people, linked to gender, occupation, sexuality, birthplace; race manually coded; SSA name data for gender.
- Embeddings: a-la-carte (conText) word embeddings + Doc2Vec page/book/image-description embeddings; normed embedding regressions test differences across collections.
- Sentiment: NRC VAD lexicon scores Valence, Arousal, Dominance of the words most associated with each gender, in both text and images.

Headline: Texas ~ California; Religious Books Stand Apart

Figure 2: Textbook Embeddings

Note: t-SNE maps of book-level TEXT (a) and IMAGE-description (b) embeddings: books cluster by subject, and within subject California and Texas overlap while Religious separates. Panels (c)-(d): normed coefficients from embedding regressions. The California-vs-Texas difference is small and NOT significant (0.04, p=0.21 text / 0.23 image), whereas Religious-vs-Texas and Religious-vs-California gaps are larger and highly significant. Contrary to the polarization narrative, the two public systems teach very similar content. (Figure 2)

Values: Religious Books Emphasize Religious & Character Values



Note: Share of pages containing character values (a) and religious values (b), by subject and collection. Character values are pervasive everywhere (Science 17-18%, Social Studies 40-48%), but Religious Reading books are highest (71%). Religious values are far more prevalent in the Religious collection across ALL subjects - 15.9% of Social Studies pages vs. 5.6% (CA) and 2.9% (TX), and 2.1% even in Science where public books are near zero. Texas and California show no meaningful difference. (Figure 3)

NEW Framework: Market & Regulatory Forces Drive Content

- Publishers serve markets; content is chosen like a product position (Hotelling/Downs). The question: why converge in public, diverge in religious?
- Public-school publishers (e.g. Pearson, McGraw Hill) serve a NATIONAL market: they build one baseline 'national' edition and only customize per state when sales justify the fixed cost F.
  - State standards restrict the feasible set; with high customization costs, the baseline converges to the 'safe union' of topics required in some state and banned in none.
  - Prediction 1 - positive SPILLOVERS: a topic required by one state appears in other states' editions too. Prediction 2 - CHILLING: topics banned by an influential state get dropped from the baseline.
- Religious publishers serve a SEGMENTED, less-regulated market not bound by state standards -> free to DIFFERENTIATE toward their target market's demographics, beliefs, and values.
- Policy implication: expanding public funds for private schooling without shared standards could increase curricular - and long-run belief - divergence.

Evidence: Standards Revisions Spill Over Across States

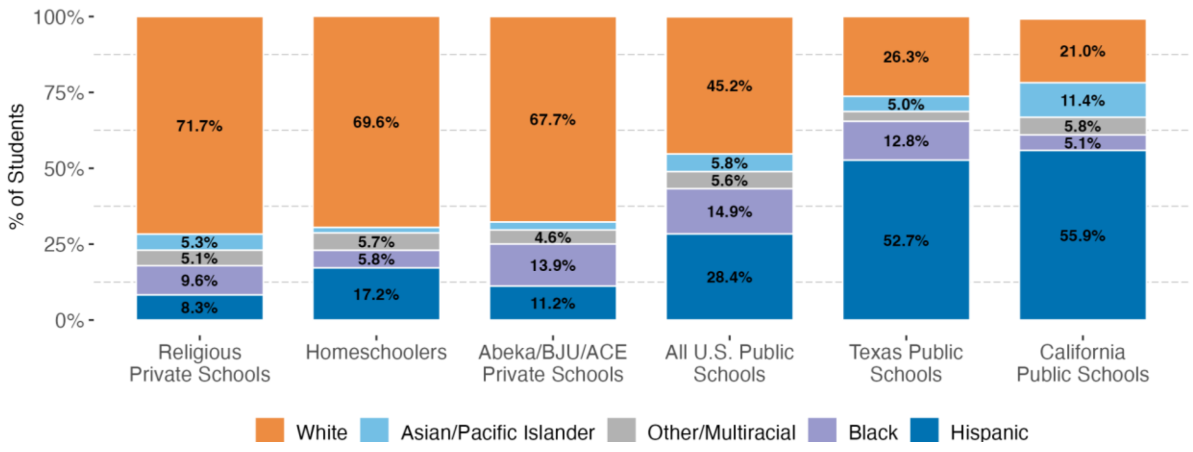
Table 2: Direct Effects and Cross-State Spillovers from State Standards Revisions

	Dependent Variable: % New-Standards Words			
	All Subjects	Reading	Science	Social Studies
	(1)	(2)	(3)	(4)
Direct Effect	0.0160***	0.0101***	0.0244***	0.0042**
	(0.0020)	(0.0018)	(0.0031)	(0.0017)
Spillover	0.0023***	0.0018*	0.0026	0.0027**
	(0.0008)	(0.0010)	(0.0021)	(0.0011)
Fixed-effects				
Collection-Grade-Subject	Yes			
Collection-Grade		Yes	Yes	Yes
Observations	735	285	271	179
Within R^2	0.49186	0.39748	0.67932	0.35463

*Note: Textbook-year regressions of the share of 'new-standards' words (words new to the revising state's standards) on post-adoption indicators, with collection-grade(-subject) fixed effects. Direct effect: the revising state's books adopt ~16 new-standards words per 1,000 (0.0160). Spillover: the OTHER state's books also rise (~2.3 per 1,000; 0.0023***) despite no change to ITS standards - positive for Reading and Social Studies. No Science spillover, consistent with divergent Science standards (California adopted NGSS in 2013; Texas did not). (Table 2)*

Differentiation Aligns with Target-Market Preferences

Figure 7: Racial/Ethnic Composition of K–12 Students by School Type



Note: Racial/ethnic composition of K-12 students by school type (NCES). Religious private, homeschool, and Abeka/BJU/ACE populations are ~68-72% White. (Figure 7)

- Religious/homeschool students are ~70% White vs. ~21% (CA) and ~26% (TX) public - a demographic gap that rationalizes lighter, whiter imagery.
- Consistent with this, Religious books depict lighter skin (even conditioning on race) and more White people in text and images.
- GSS beliefs of religious respondents track the content: more rejection of evolution, support for school prayer, opposition to homosexuality, and traditional gender norms (Fig. 8).
- Portrayal (VAD, Fig. 6): across ALL collections females are framed more positively but less active and less powerful than males; Religious books have the lowest

Takeaways

- No deep public-school divide: despite the 'red vs. blue' narrative, Texas and California textbooks are strikingly similar in topics, values, portrayal, and representation.
- Church vs. state IS visible: religious books emphasize religious and character values, show less female representation, and depict lighter-skinned, more White figures - including in Science.
- A market explanation: state standards + customization costs push public publishers toward a converged national baseline (with measurable cross-state spillovers); unregulated religious publishers differentiate toward their target market.
- Shared blind spots everywhere: minimal LGBTQIA+, Asian, Latine, and Indigenous representation; females portrayed positively but less powerfully.
- Methodological contribution: a scalable AI + text/image pipeline (LMM, LLM, computer-vision skin analysis, NER, embeddings, VAD) for measuring presence and portrayal in any large document corpus.